



LINEAR ALGEBRA AND ITS APPLICATIONS

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MATRICES AND GAUSSIAN ELIMINATION

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Course Content: Inverses and Transposes

- ❖ Let A be a square matrix of order n the **Inverse of A** is the matrix B such that **$AB=I=BA$** .
Here $B=A^{-1}$.
- **Properties:** Inverse of a matrix is unique. i.e., $AB=BA=I$ and $AC=CA=I$, then $B=C$
- Inverse of the product is the product of Inverses. $(ABCD)^{-1}=D^{-1} C^{-1} B^{-1} A^{-1}$
- If $A=LU$ then $A^{-1}=U^{-1} L^{-1}$.
- Since $E_{32}E_{31}E_{21}A=U$ we have $E_{21}^{-1}E_{31}^{-1}E_{32}^{-1}U=A \Rightarrow A^{-1}=U^{-1}E_{32}E_{31}E_{21}$
$$\Rightarrow L=E_{21}^{-1}E_{31}^{-1}E_{32}^{-1}$$
- A matrix A is invertible if and only if elimination produces n pivots with or without row exchanges. Elimination solves $Ax=b$ without explicitly finding A^{-1} .
- If A is invertible, the one and only one solution to $Ax=b$ is $x=A^{-1}b$

Gauss Jordan Method of computing A^{-1} :

- ❖ The inverse of an invertible matrix is obtained by a set of row operations that transforms A to I and I to A^{-1} . This process is known as **Gauss Jordan Method**.
- ❖ Consider the Augmented Matrix **$[A:I]$** . Then perform row operations on it so that A reduces to Echelon form **U** and at the same time **I** reduces to **C** . Further reduce **U to I** using elementary row transformations which reduces **C to A^{-1}** .

❖ i.e., $[A:I] \rightarrow [U:C] \rightarrow [I:A^{-1}]$

Ex: Compute A^{-1} using Gauss Jordan Method given

$$A = \begin{pmatrix} 2 & 1 & 1 \\ 4 & 3 & 4 \\ -2 & 2 & 2 \end{pmatrix}$$

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$$\begin{aligned} \text{Ex: } [A:I] &= \begin{pmatrix} 2 & 1 & 1:1 & 0 & 0 \\ 4 & 3 & 4:0 & 1 & 0 \\ -2 & 2 & 2:0 & 0 & 1 \end{pmatrix} \xrightarrow[R_3 + R_1]{R_2 - 2R_1} \begin{pmatrix} 2 & 1 & 1:1 & 0 & 0 \\ 0 & 1 & 2:-2 & 1 & 0 \\ 0 & 3 & 3:1 & 0 & 1 \end{pmatrix} \\ &\xrightarrow{R_3 - 3R_2} \begin{pmatrix} 2 & 1 & 1:1 & 0 & 0 \\ 0 & 1 & 2:-2 & 1 & 0 \\ 0 & 0 & -3:7 & -3 & 1 \end{pmatrix} = [U:C] \xrightarrow[R_2 + 2/3R_3]{R_1 + 1/3R_3} \begin{pmatrix} 2 & 1 & 0:10/3 & -1 & 1/3 \\ 0 & 1 & 0:8/3 & -1 & 2/3 \\ 0 & 0 & -3:7 & -3 & 1 \end{pmatrix} \\ &\xrightarrow{R_1 - R_2} \begin{pmatrix} 2 & 0 & 0:2/3 & 0 & -1/3 \\ 0 & 1 & 0:8/3 & -1 & 2/3 \\ 0 & 0 & -3:7 & -3 & 1 \end{pmatrix} \xrightarrow[R_3 = -1/3R_3]{R_1 = 1/2R_1} \begin{pmatrix} 1 & 0 & 0:1/3 & 0 & -1/6 \\ 0 & 1 & 0:8/3 & -1 & 2/3 \\ 0 & 0 & 1:-7/3 & 1 & -1/3 \end{pmatrix} \\ &= [I:B] \\ B = A^{-1} &= \begin{pmatrix} 1/3 & 0 & -1/6 \\ 8/3 & -1 & 2/3 \\ -7/3 & 1 & -1/3 \end{pmatrix} \end{aligned}$$

Transpose of a Matrix A^T :

❖ If $A = [a_{ij}]_{m \times n}$ is an $m \times n$ matrix, then its transpose is obtained by interchanging its rows and columns and is denoted by $A^T = [a_{ji}]_{n \times m}$.

Ex: If $A = \begin{pmatrix} 2 & 1 & -3 \\ 4 & 2 & 0 \end{pmatrix}_{2 \times 3}$ then $A^T = \begin{pmatrix} 2 & 4 \\ 1 & 2 \\ -3 & 0 \end{pmatrix}_{3 \times 2}$

➤ Properties:

➤ The Transpose of a Lower Triangular Matrix is an Upper Triangular Matrix.

$$(A^T)^T = A ; (AB)^T = B^T A^T ; (A^{-1})^T = (A^T)^{-1} ;$$

$$(A \pm B)^T = A^T \pm B^T ; (A^{-1})^T A^T = (AA^{-1})^T = I$$

Symmetric Matrices:

❖ If A is a matrix of order n is said to be symmetric matrix of order if $A^T = A$

Ex: If $A = \begin{pmatrix} 2 & -3 \\ -3 & 5 \end{pmatrix}$ then $A^T = \begin{pmatrix} 2 & -3 \\ -3 & 5 \end{pmatrix}$

➤ Properties:

- If A is symmetric then A^{-1} may or may not exist.
- If for a symmetric matrix A^{-1} exists then A^{-1} is also symmetric.
- For a symmetric matrix A , we have $(A^{-1})^T = (A)^{-1}$ ($\because A^T = A$)

Symmetric Products AA^T , $A^T A$, LDL^T :

❖ If A is a matrix of order $m \times n$, then AA^T and $A^T A$ are both symmetric

Ex: If $A = \begin{pmatrix} 2 & 1 & -3 \\ 4 & 2 & 0 \end{pmatrix}_{2 \times 3}$ then $AA^T = \begin{pmatrix} 14 & 10 \\ 10 & 20 \end{pmatrix}$ $A^T A = \begin{pmatrix} 20 & 10 & -6 \\ 10 & 5 & -3 \\ -6 & -3 & 9 \end{pmatrix}$

➤ If A is symmetric and if $A = LDU$ then,

$$A = A^T = LDL^T \quad (\because U = L^T \text{ \& } L = U^T)$$

$$A = \begin{pmatrix} 1 & 2 & -1 \\ 2 & 3 & 0 \\ -1 & 0 & 4 \end{pmatrix} \xrightarrow[R_3 + R_1]{R_2 - 2R_1} \begin{pmatrix} 1 & 2 & -1 \\ 0 & -1 & 2 \\ 0 & 2 & 3 \end{pmatrix} \xrightarrow{R_3 + 2R_2} \begin{pmatrix} 1 & 2 & -1 \\ 0 & -1 & 2 \\ 0 & 0 & 7 \end{pmatrix} = U$$

$$LDU = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & -2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 7 \end{pmatrix} \begin{pmatrix} 1 & 2 & -1 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{pmatrix} \quad U = L^T$$

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MATRICES AND GAUSSIAN ELIMINATION:

❖ For which three numbers “c” is this matrix not invertible, and why not?

$$A = \begin{pmatrix} 2 & c & c \\ c & c & c \\ 8 & 7 & c \end{pmatrix} \xrightarrow[R_3 - 4R_1]{R_2 - c/2R_1} \begin{pmatrix} 2 & c & c \\ 0 & c - (c^2/2) & c - (c^2/2) \\ 0 & 7 - 4c & -3c \end{pmatrix}$$
$$\xrightarrow{R_3 - \left(\frac{7 - 4c}{2c - c^2}\right)2R_2} \begin{pmatrix} 2 & c & c \\ 0 & c - (c^2/2) & c - (c^2/2) \\ 0 & 0 & c - 7 \end{pmatrix}$$

- Matrix A is not invertible for $c=0,2,7$
- For $c=0,2,7$ elimination gives one zero row, hence A will be singular and so A will not be invertible.



THANK YOU

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